
Container Closure Systems for Human Drugs and Biological Products

Guidance for Industry

DRAFT GUIDANCE

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Center for Drug Evaluation and Research (CDER)
Center for Biologics Evaluation and Research (CBER)
Office of Combination Products (OCP)**

**August 2026
Pharmaceutical Quality/CMC**

Container Closure Systems for Human Drugs and Biological Products Guidance for Industry

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Container Closure Systems for Human Drugs and Biological Products

Guidance for Industry¹

This draft guidance, when finalized, will represent the current thinking of the Food and Drug Administration (FDA or Agency) on this topic. It does not establish any rights for any person and is not binding on FDA or the public. You can use an alternative approach if it satisfies the requirements of the applicable statutes and regulations. To discuss an alternative approach, contact the FDA staff responsible for this guidance as listed on the title page.

I. INTRODUCTION

This document provides guiding principles² for evaluating the quality of **container closure systems (CCSs)**³ used to package drugs and biological products, for human use. This includes CCSs that are also device constituent parts of **combination** products and CCSs used to package the drug or biological product constituent parts of combination products.⁴ Specifically, this guidance discusses FDA's current thinking on how to evaluate CCSs according to a risk-based framework that includes quality assessments and quality control of packaging materials and components.⁵ When final, this guidance will supersede the guidances for industry *Container Closure Systems for Packaging Human Drugs And Biologics* (May 1999) and *Container Closure Systems for Packaging Human Drugs and Biologics — Questions and Answers* (May 2002).⁶ FDA intends to supplement this guidance with additional guidances to provide topic-specific recommendations related to the general information described in this guidance. These topic-specific guidances will address methods related to evaluating and characterizing novel CCSs and provide information about specific quality attributes and testing requirements, including

¹ This guidance has been prepared by the Office of Pharmaceutical Quality in the Center for Drug Evaluation and Research in cooperation with the Center for Biologics Evaluation and Research, Center for Devices and Radiological Health, Office of Regulatory Affairs (now known as the Office of Inspections and Investigations), and Office of Combination Products in the Office of the Commissioner at the Food and Drug Administration.

² This guidance does not suggest specific test methods or acceptance criteria (except for references to United States Pharmacopeia methods), nor does it suggest comprehensive lists of tests. These details should be determined based on good scientific principles for each specific container closure system for specific drug product formulations, dosage forms, and routes of administration. Acceptance criteria should be based on data for specific packaging components and container closure systems.

³ Terms bolded at first mention are defined in the Glossary.

⁴ See 21 CFR 3.2(e) and 21 CFR 4.2. If you have questions regarding whether a CCS is a device and your product is, therefore, a combination product, contact the Office of Combination Products at combination@fda.gov for assistance.

⁵ FDA responds to potential drug shortages by taking actions to address their underlying causes and enhance product availability, as appropriate. When drug shortages are caused by quality problems, including problems involving a CCS, FDA works with the firm to address the issues. For drug shortage information and reporting, see <https://www.fda.gov/drugs/drug-shortages/frequently-asked-questions-about-drug-shortages>.

⁶ We update guidances periodically. For the most recent version of a guidance, check the FDA guidance web page at <https://www.fda.gov/regulatory-information/search-fda-guidance-documents>.

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considerations for **extractables** and **leachables** evaluations and associated toxicological risk assessments.

In general, FDA’s guidance documents do not establish legally enforceable responsibilities. Instead, guidances describe the Agency’s current thinking on a topic and should be viewed only as recommendations, unless specific regulatory or statutory requirements are cited. The use of the word *should* in Agency guidances means that something is suggested or recommended, but not required.

II. SCOPE

This guidance provides recommendations regarding pharmaceutical quality considerations (e.g., chemistry, manufacturing, and controls (CMC)) for CCSs used for human drugs and biological products.⁷ It discusses both general principles and specific considerations for the development and assessment of CCSs for drug products,⁸ and drug substances. It also applies to CCSs that are device constituent parts of combination products or that are used to hold the drug or biological product constituent part(s) of these products.⁹ Although the scope of this guidance excludes stand-alone devices, and its focus is on CCSs that package drugs or biological products, some recommendations described in this guidance also may be applicable to stand-alone devices intended to deliver drugs.¹⁰

This guidance applies to applications¹¹ including amendments and supplements to approved applications, for human drugs and biological products, as well as for combination products. It also applies to drug products legally marketed according to section 505G of the Federal Food,

⁷ For the purposes of this guidance, the term *drug* refers to drug substance and drug product, including biological products. *Biological products* refer to those defined in section 351(i)(1) of the Public Health Service Act and subject to licensure under section 351(a) or (k) of the Act. In this guidance, *drug* also includes drug or biological product constituent parts of combination products.

⁸ For the purposes of this guidance, *drug product* refers to any drug in finished dosage form.

⁹ In accordance with 21 CFR 3.4, a combination product is assigned to an Agency center that will have primary jurisdiction for that combination product’s premarket review and regulation. A drug or biological product constituent part may be part of a CDER-, CBER-, or CDRH-led combination product. Combination products composed of a device constituent part that is the CCS for a drug or biological product constituent part are typically assigned to CDER or CBER.

¹⁰ Questions regarding whether these recommendations would apply to a specific device should be discussed with the CDRH or CBER review division during the development process. In this guidance, the term *device* has the meaning given in section 201(h) of the Federal Food, Drug, and Cosmetic Act (21 U.S.C. 321(h)).

¹¹ For the purposes of this guidance, the term *application* refers to investigational new drug applications, new drug applications, abbreviated new drug applications, and biologics license applications. It also includes investigational device exemption applications, premarket approval applications, De Novo requests, and premarket notifications (as described in sections 515, 513(f)(2), and 510(k) of the Federal Food, Drug, and Cosmetic Act (21 U.S.C. 360e, 360c(f)(2), and 360(k)), respectively) for combination products covered by this guidance.

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Drug, and Cosmetic Act (FD&C Act)¹² without an approved application under section 505 of the FD&C Act.^{13,14}

This guidance is not intended to address packaging operations (e.g., processes of filling, packaging, and labeling) associated with drug product manufacture.

This guidance does not cover standards and recommendations for child-resistant packaging. Regulations issued under the Poison Prevention Packaging Act of 1970 (PPPA)¹⁵ establish the standards and test procedures for child-resistant packaging.¹⁶ For applications that propose child-resistant packaging statements in product labeling, the applicant should follow the recommendations in the guidance for industry *Child-Resistant Packaging Statements in Drug Product Labeling* (August 2019), which include submitting written verification to FDA that the drug product's child-resistant packaging meets the standards under 16 CFR part 1700.

III. BACKGROUND

A CCS is the sum of **packaging components** that together contain and protect a drug. It includes the **primary packaging components**, and it also includes the **secondary packaging components** if their purpose is to further protect the drug. A CCS must provide adequate protection in storage and use without altering the safety, identity, strength, quality, or purity of the drug beyond the official or established requirements.¹⁷ For example, packaging components should be constructed of materials that will not leach harmful or undesirable amounts of substances to which patients could be exposed directly or indirectly.

In recent years, the development of novel drug products and new container closure technologies has led to advancements in the design, construction, and evaluation of CCSs. For example, technological advancements in pharmaceutical therapies can introduce additional functional purposes for CCSs, as with products that use the CCS to prepare the final dosage or to deliver the product. In such cases, additional control measures and testing should be conducted to ensure that the additional functions of the CCS do not adversely impact the safety, identity, strength, quality, and purity of the product.

To accommodate the advancements in the design, construction, and evaluation of CCSs, several

¹² See 21 U.S.C. 355h.

¹³ See 21 U.S.C. 355.

¹⁴ Drugs marketed pursuant to section 505G of the FD&C Act must also meet the general requirements for nonprescription drugs, which include requirements that these drugs contain only safe and suitable inactive ingredients and are manufactured according to current good manufacturing practice. It is the manufacturer's responsibility to comply with these requirements and to document and maintain supporting data in its files.

¹⁵ Authority under the PPPA is vested in the Consumer Product Safety Commission by section 30(a) of the Consumer Product Safety Act. Special packaging is defined under 15 U.S.C. 1471(4), 16 CFR 1700.1(b)(4), and 21 CFR 310.3(l). Regulations issued under the PPPA establish performance standards and test methods that determine if a packaging system is child-resistant and adult-use-effective (16 CFR 1700.15 and 16 CFR 1700.20). The written verification to FDA discussed in this guidance is separate from the certification required to be provided to the Consumer Product Safety Commission under 15 U.S.C. 2063 and 16 CFR 1110.

¹⁶ See 16 CFR part 1700.

¹⁷ See 21 CFR 211.84 and 211.94.

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general chapters within the United States Pharmacopeia and the National Formulary (USP-NF) provide new or updated information relevant to the storage and distribution of pharmaceutical products, including more comprehensive and advanced testing to characterize packaging materials and components. Other FDA and International Conference on Harmonization (ICH) guidances^{18,19} that outline specific recommendations relevant to CCS assessments have also been issued since the 1999 publication of the guidance for industry *Container Closure Systems for Packaging Human Drugs and Biologics*.

A. Statutory and Regulatory Authority

The FD&C Act contains packaging requirements for drug products. A drug is deemed adulterated “if its container is composed, in whole or in part, of any poisonous or deleterious substance which may render the contents injurious to health” (see section 501(a)(3)²⁰). A drug is also deemed adulterated if the methods, facilities, or controls used for its manufacture, processing, packing, or holding do not conform to or are not operated or administered according to current good manufacturing practice (CGMP, see section 501(a)(2)(B)²¹). Additionally, under section 502,²² a drug is deemed misbranded if its packaging, including labeling, does not meet other requirements set forth in the Act. An application must include a description of the manufacturing and packing procedures as described in 21 CFR 314.50(d)(1) for new drug applications, in 21 CFR 314.94(a)(9) for abbreviated new drug applications, and in 21 CFR 601.2 for biologics license applications. For a device-led combination product reviewed in an appropriate device application, such a description as noted is among the types of information that would typically be necessary to support the safety and effectiveness of the combination product.²³ For all drug products and combination products, information to demonstrate compliance with CGMP with respect to the CCS (e.g., 21 CFR 4, subpart A, 21 CFR 211 subpart E, 21 CFR part 820)²⁴ must be available on inspection or upon a request under section 704(a)(4) of the FD&C Act.

¹⁸ See, for example, the guidance for industry *Nasal Spray and Inhalation Solution, Suspension, and Spray Drug Products — Chemistry, Manufacturing, and Controls Documentation* (July 2002). See also the draft guidance for industry *Transdermal and Topical Delivery Systems — Product Development and Quality Considerations* (November 2019). When final, this guidance will represent FDA’s current thinking on this topic.

¹⁹ See, for example, the ICH guidance for industry *Q8(R2) Pharmaceutical Development* (November 2009).

²⁰ See 21 U.S.C. 351(a)(3).

²¹ See 21 U.S.C. 351(a)(2)(B).

²² See 21 U.S.C. 352.

²³ As explained in the guidance for industry and FDA staff *Principles of Premarket Pathways for Combination Products* (January 2022), in determining what is needed to demonstrate the safety and effectiveness of the combination product, FDA takes into account the questions and considerations, reflected in the statutory and regulatory provisions associated with each constituent part in its review of the combination product as a whole and its constituent parts. This includes how the constituent parts may interact and interrelate. For example, for a device-led combination product reviewed in an appropriate device application that includes a drug constituent part (that would otherwise be reviewed in an NDA), nonclinical pharmacology and toxicology and clinical pharmacology (including pharmacokinetic) data and chemistry, manufacturing, and controls (CMC) information are among the types of information that would typically be necessary.

²⁴ On February 2, 2024, FDA issued a final rule amending the device quality system regulation, 21 CFR part 820, to align more closely with international consensus standards for devices and conforming amendments were also made to 21 CFR part 4 (89 FR 7496, available at <https://www.federalregister.gov/d/2024-01709>). This final rule took

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B. Compendial Requirements

Section 201(j) of the FD&C Act²⁵ recognizes the USP-NF as an official compendium of the United States. Under section 501(b) of the FD&C Act,²⁶ a drug is deemed to be adulterated if it purports to be or is represented as a drug the name of which is recognized in the USP-NF and its strength differs from, or its quality and purity falls below, the standards set forth in the compendium. Under section 502(g) of the FD&C Act,²⁷ a drug with a name recognized in the USP-NF is deemed misbranded unless it is packaged and labeled as described in the compendium. Standards for an article recognized in a USP-NF official monograph, general chapter, or general notice apply for the article's entire life cycle (production to expiration). Various USP general chapters define requirements for **materials of construction (MOC)**, components, and CCSs. For new or newly revised standards, USP allows, and FDA generally encourages, early adoption of revised standards in advance of the official date unless otherwise specified at the time of publication.²⁸

C. Combination Products

A combination product is a human medical product composed of two or more types of medical products (i.e., drug and device, drug and biological product, device and biological product, or all three together).²⁹ The drugs, devices, and biological products included in combination products are referred to as *constituent parts* of the combination product. A CCS may also be a device constituent part if it has more than one function (e.g., to be a container closure and to deliver the drug). Examples of CCSs that may also be a device constituent part of a combination product include piston syringes, pumps, metered dose inhalers (MDIs), and intravenous bags. Combination products raise additional premarket review considerations relating, for example, to performance of the device, engineering, and user interface, all of which may be affected by interactions (e.g., adsorption) between the device and drug or biological product it holds.³⁰ Combination products are also subject to additional postmarket regulatory requirements, including 21 CFR part 4 subparts A and B.³¹ Some considerations for CCSs that are device constituent parts are addressed in this guidance (see section V.B.3). However, sponsors should

effect on February 2, 2026. This rule revised or removed certain provisions in part 820 and incorporated by reference the 2016 edition of the International Organization for Standardization (ISO) 13485, Medical devices – Quality management systems – Requirements for regulatory purposes, in part 820. As stated in the preamble to the final rule, the requirements in ISO 13485 are, when taken in totality, substantially similar to the requirements of the current part 820, providing a similar level of assurance in a firm's quality management system and ability to consistently manufacture devices that are safe and effective and otherwise in compliance with the Federal Food, Drug, and Cosmetic Act (FD&C Act). See 89 FR 7496, available at <https://www.federalregister.gov/d/2024-01709>.

²⁵ See 21 U.S.C. 321(j).

²⁶ See 21 U.S.C. 351(b).

²⁷ See 21 U.S.C. 352(g).

²⁸ See USP General Notices and Requirements <3.10> *Applicability of Standards* for USP early adoption policies at the time of reference.

²⁹ See footnotes 4 and 9.

³⁰ For further information, see the guidance for industry and FDA staff, *Principles of Premarket Pathways for Combination Products* (January 2022).

³¹ For further information, see the guidance for industry and FDA staff, *Current Good Manufacturing Practice Requirements for Combination Products* (January 2017), and *Postmarketing Safety Reporting for Combination Products* (July 2019).

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refer to other relevant guidance including those cited in this guidance, and they may contact the review division for their combination product with questions regarding data and information needed to support marketing authorization.

IV. RISK-BASED FRAMEWORK: FACTORS TO CONSIDER

Selection of the CCS for a drug depends on the characteristics of both the drug and the CCS. It cannot be assumed that because a CCS was suitable for one drug product, it is automatically appropriate for another product. A CCS's **critical quality attributes**,³² particularly its leachables profile, can be affected by differences in manufacturing processes, variations in materials suppliers, or even subtle changes in the drug product's formulation.

A suitable framework for evaluating the quality of a CCS should use a risk management process.³³ The process should begin with an assessment of the CCS used for a specific drug, including the CCS's selected MOC, the drug's manufacturing processes, and the drug's clinical use. The potential risks associated with the CCS should be adequately identified and characterized based on these assessments. For example, interactions between drugs and packaging materials could produce leachables that cause unintended chemical toxicity. Or, these leachables could cause undesirable changes in the drug's quality (e.g., reduced stability, variations in the impurity profile, increased particulate matter, altered antigenicity) that result in an unacceptable alteration of the drug's intended therapeutic effects. The qualification and quality control of packaging components should include mitigation strategies that address knowledge gaps and risks identified through testing and other evaluations.

A. General Considerations

The following factors should be considered when using a risk-based approach to evaluate a CCS. This list is not exhaustive; additional risk factors may need to be evaluated during the product's development and life cycle.

1. Safety of Packaging Materials

Packaging components should be constructed of materials that will not leach harmful or undesirable amounts of substances to which a patient will be exposed when being treated with the drug. The primary CCS is the most likely source of leachables because the packaging components are in direct contact with the drug, but leachables can also come from secondary CCSs (e.g., protective packaging, inks, labels, adhesives) and from manufacturing equipment used in drug production. Assessments of safety- and compatibility-related risks should be

³² A critical quality attribute is a physical, chemical, biological, or microbiological property or characteristic that should be within an appropriate limit, range, or distribution to ensure the desired product quality. See the ICH guidance for industry *Q8(R2) Pharmaceutical Development* (November 2009).

³³ See the ICH guidance for industry *Q9(R1) Quality Risk Management* (May 2023) and FDA draft guidance, *Benefit-Risk Considerations for Product Quality Assessments* (May 2022). When final, this guidance will represent FDA's current thinking on this topic.

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relevant to the drug's intended use (e.g., route of administration, delivery mechanism) and the dosage form (e.g., solid- or liquid-based).

Containers and closures must not interact with a drug product to cause unacceptable changes in the quality of either the drug product or the packaging components.³⁴ Similarly, packaging components should not interact with the drug substance to compromise quality. Examples of adverse outcomes caused by incompatibility between a CCS and a drug include:

- Increased degradation of the active pharmaceutical ingredient
- Decreased effectiveness of key excipients (e.g., antioxidants or preservatives)
- Changes in physicochemical properties of drugs (e.g., precipitation, pH, appearance)
- Compromise of packaging materials or their performance (e.g., glass delamination, formation of particulates)
- Change in product stability caused by interactions between a packaging component and a drug (e.g., the types and amounts of leachables could increase over storage time as a result of such interactions)
- Change in a biological product's effectiveness and safety caused by leachables interacting or reacting with a drug that alters its properties (e.g., reacting with a protein and causing changes to its tertiary structure)

2. Protection

A CCS must protect against ingress from external contaminants and should prevent leakage.³⁵ A CCS must also protect against external factors that can cause deterioration or contamination of the drug product,³⁶ such as exposure to light, loss of solvent, physical stress during transportation and storage, exposure to reactive gases (e.g., oxygen), absorption of water vapor, and ingress of microbial contaminants.

3. Performance

A CCS's performance is its ability to function in the manner for which it was designed. A CCS can have functional features in addition to containing and protecting drugs. For example, for a multiple-dose product, the elastomeric closures of the CCS are pierced multiple times; it is important to evaluate the performance of the closure during the piercing process. Specifically, the functional parameters that include penetrability, fragmentation, and self-sealing efficiency should be evaluated.³⁷ As another example, an MDI CCS consists of the device constituent part, which plays an important role in generating aerosol particles, determining the aerosol characteristics, and controlling the amount of medication delivered to the patient. Tests to characterize the delivery performance are an important part of evaluating an MDI.³⁸ The risk

³⁴ See 21 CFR 211.84 and 211.94.

³⁵ See 21 CFR 211.94.

³⁶ Ibid.

³⁷ See USP General Chapter <381> *Elastomeric Components in Injectable Pharmaceutical Product Packaging/Delivery Systems*.

³⁸ See the draft guidance for industry *Metered Dose Inhaler (MDI) and Dry Powder Inhaler (DPI) Drug Products — Quality Considerations* (April 2018). When final, this guidance will represent FDA's current thinking on this topic.

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assessment should account for the functionality of the CCS and discuss the mitigations for the risks that may prevent the CCS from performing those functions.

A CCS that performs drug delivery should deliver the drug to the intended delivery site in the amount and rate described in the drug labeling.³⁹ As appropriate, data should be provided to demonstrate that the CCS operates as designed (see sections III.C and V.B.4 for additional information).

Integrity failures in primary packaging components could cause dimensional incompatibility among the packaging components (e.g., closures with imperfect seals, plunger-syringe incompatibility, variation in dimensional parameters between stopper and container neck). These incompatibilities can further affect CCS performance.⁴⁰

4. Impacts of Manufacturing Process

The risk assessment should include an evaluation of any potential effects of direct or indirect treatment (e.g., washing, coating, lyophilization, sterilization) of the CCS during the drug manufacturing process. The compatibility of a CCS with a given treatment (a pre- or post-processing method) should be evaluated to ensure that these processes will not adversely affect the CCS's suitability, compatibility, or functionality. Terminal sterilization and depyrogenation processes that employ high heat, irradiation, or exposure to reactive gases (e.g., ethylene oxide, vaporized hydrogen peroxide) could negatively affect the CCS's quality. For example, these processes could cause glass to delaminate, which could result in particulate matter in the drug product, and insufficient aeration during sterilization by vapor hydrogen peroxide or ethylene oxide, which could cause leaching of harmful chemicals from the packaging components to the drug.

5. Storage and Handling

The risk assessment should account for the product's intended storage and handling conditions to ensure that the CCS can maintain integrity and functional performance. For example, if the final product is stored at very low temperatures (e.g., frozen temperature of -20°C or cryotemperature of less than -160°C), the integrity and functionality of a CCS should be assessed after storage under worst-case conditions. One way to conduct this type of assessment is to use freeze-thaw cycles or thermal cycling to simulate the storage and handling of the product (see section V.A.5).

B. Risk Considerations for Common Dosage Forms of Drug Products

The Agency recognizes that certain drug products pose higher risk than others. The type and extent of information (e.g., testing data) that should be provided for assessments of a CCS is based on the risk identified. Because the level of risk associated with a CCS depends on multiple

³⁹ See draft guidance for industry *Essential Drug Delivery Outputs for Devices Intended to Deliver Drugs and Biological Products* (June 2024). When final, this guidance will represent FDA's current thinking on this topic.

⁴⁰ If a CCS integrity failure is detected, manufacturers must investigate the root cause. Based on a robust quality risk assessment, manufacturers must also implement appropriate corrective actions, which may include changes to the manufacturing process to reduce or prevent further CCS integrity failures (see 21 CFR 211.22(a) and 211.192).

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and complex factors, risk assessments should be product-specific. For example, an assessment for leachables should account for the safety of the CCS's MOC, the likelihood that leachables would release into the product from the CCS, and the potential toxicity of the leachables.

Table 1 provides examples of packaging concerns for common dosage forms of drug products categorized by the route of administration (ROA) and the likelihood of interaction between the packaging components and the drug product. The columns reflect the relative degree of concern associated with each listed ROA, and the rows reflect the likelihood of interaction between the packaging components and drug product. This table is meant to illustrate that not all CCSs have the same degree of packaging concerns; however, it should not be viewed as justification for not adequately assessing the safety of a particular proposed CCS. CCSs with higher risk generally require greater scrutiny to ensure safe use. However, even low-risk drug products present some risk and an adequate assessment for potential leachables is important for these particular drug products (e.g., topical and oral dosage forms).

Table 1 is not a complete list of all possible dosage forms and ROAs, but the general principles used to determine risk levels in this table apply to other dosage forms and ROAs.. The risk assessments and extent or adequacy of recommended testing of a CCS for any given product should not be determined based solely on the examples in the table. All risk factors for a specific product should be taken into consideration (e.g., dosage forms, CCS MOC and design, drug formulation, ROA) in the overall risk evaluation of a CCS. Factors that are not specified in the table can significantly affect the level of concern, which may differ even among drug products with the same ROA or dosage forms. For example, drug formulation information including cosolvent, physicochemical properties of drug substance and excipients, drug load, pH, surfactants, chelating agents, preservatives, stabilizers, and lipids should be considered in the risk assessment as applicable, along with any additional relevant factors that may affect risk. Risk considerations for common dosage forms of drug products are further discussed in detail following the table.

Table 1. Examples of Packaging Concerns for Common Dosage Forms of Drug Products^a

Degree of concern associated with the route of administration ^b	Likelihood of interaction between the packaging components and dosage form ^b		
	High	Medium	Low
Highest	Inhalation aerosols (propellant based)	Injectable solutions and suspensions Inhalation solutions and suspensions ^c	Inhalation powders Powders for injection
High	Transdermal products and delivery systems Nasal aerosols (propellant-based)	Ophthalmic solutions and suspensions Nasal sprays Otic solutions and suspensions	—

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Medium	Topical dermal products (liquid and semisolid)	Lingual and buccal sprays	Buccal/sublingual tablets Topical powders
Low ^d	Oral solutions and suspensions (liquid and semisolid)	—	Oral tablets and capsules Oral powders

^a This table provides examples of the general level of packaging concern for several common dosage forms of drug products. The risk associated with a CCS and the extent or adequacy of recommended testing for any given product should not be determined solely based on the examples provided in the table. For combination products, the risk and extent and adequacy of testing should also take into account whether interactions between a CCS that is a device constituent part and the drug or biological product constituent part it holds (such as adsorption of the drug or biological product constituent part by the CCS) might negatively affect device performance.

^b Factors that are not specified in the table, such as drug formulation composition (e.g., aqueous vs. nonaqueous), can significantly affect the level of concern even compared to other drug products with the same ROA or dosage form. All risk factors (e.g., dosage forms, CCS material and design, drug formulation) should be considered based on the specific product in the overall risk evaluation of a CCS.

^c The term *suspension* is used here to describe a mixture of two immiscible phases (e.g., solid in liquid, liquid in liquid). It encompasses a wide variety of dosage forms, including creams, ointments, gels, and emulsions.

^d Low-risk dosage forms are not free of risk. Though interaction of these dosage forms with packaging components is less likely than for the other listed forms, risks should be assessed and controlled as appropriate.

1. Inhalation Products

Inhalation products have one of the highest degrees of concern based on ROA because patients with compromised lung function (e.g., asthma, chronic obstructive pulmonary disease) can inhale leachables from packaging materials directly into their lungs. However, the risk levels concerning product interaction with the CCS vary based on the products' dosage forms.

MDI drug formulations contain drug substances (dissolved or suspended) in a propellant; a mixture of propellants; or a mixture of solvents, propellants, or other excipients. The CCS for an MDI consists of the device constituent parts of the combination product (i.e., the canister, the actuator, and the metering valve), including any additional features (e.g., integrated spacer, integrated dose counter),⁴¹ and can also include protective secondary packaging. Device constituent parts and their materials that may come into contact with the drug constituent part should also be considered in the risk assessment. MDIs have the highest risk of interactions between packaging components and the drug because they include a propellant. The presence of an organic propellant in MDI formulations could enhance leaching of compounds from the valve or canister components into the formulation.

Inhalation solutions and suspensions are intended for delivery to the lungs by oral inhalation for local and/or systemic effects and are used with a nebulizer. Inhalation solutions and sprays are typically aqueous-based formulations and have a medium risk of interactions between packaging components and the drug. Unit-dose presentation is recommended for these drug products to prevent microbial contamination during use. The CCS for these products can also include protective packaging such as foil overwrap (see section V.B.1).

⁴¹ See the draft guidance for industry *Metered Dose Inhaler (MDI) and Dry Powder Inhaler (DPI) Drug Products — Quality Considerations* (April 2018). When final, this guidance will represent FDA's current thinking on this topic.

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Dry powder inhaler (DPI) drug formulations typically contain the drug substance and excipients, including a drug carrier (e.g., lactose) that improves the flow of the drug. Because the drug product is a solid powder, this dosage form has a lower risk of interactions between packaging components and the dosage form compared to liquid formulations.

2. Intranasal Products

Nasal sprays are applied to the nasal cavity for local and/or systemic effects. Nasal spray combination products contain drug substances dissolved or suspended in solutions or mixtures of excipients (e.g., preservatives, viscosity modifiers, emulsifiers, buffering agents) in nonpressurized dispensers that deliver a spray containing a metered dose of the drug substance. The formulation and the CCS (container, closure, pump, and any protective packaging) collectively constitute the product. The drug formulations are typically aqueous-based and have a medium risk of interactions between packaging components and the drug.

Propellant-based nasal aerosols (nasal MDIs) have a high risk of interactions between packaging components and the drug. However, these products are not included in the highest risk portion of the table because (unlike inhalation MDIs) they are not intended for delivery to the lungs.

3. Injectable Drug Products

An injectable drug is intended for injection through the skin or other external boundary tissue (rather than for ingestion through the alimentary canal) so that drug substances are administered directly into a blood vessel, organ, tissue, or lesion. There are many routes for administration of injectable drug products (e.g., subcutaneous, intramuscular, intravenous). Injectable drug products may be liquids in the form of solutions, emulsions, or suspensions, or they may be dry solids that are to be combined with an appropriate vehicle to yield a solution or suspension. These products typically have a medium risk of interactions between packaging components and the drug product. However, the risk can be elevated for certain injectable products if the excipients used in the drug product promote the leaching of chemicals from the CCS. Injectable products that contain organic solvents (e.g., ethanol or dimethyl sulfoxide in a depot injection drug product) in their formulations have high potential to interact with their packaging materials. Drug products administered by the intravenous route are 100% bioavailable; therefore, any leachables in the product would be administered with the intended dosage form and have wide access to organs and tissues. Similar concerns exist for leachables in drug products administered by the subcutaneous and intramuscular routes, given the potential for significant exposure to organs and tissues. For injectable products that are delivered into specific tissues (e.g., ocular, perineural, intra-articular), local tissue toxicity should be considered in the safety evaluation. Drug products intended for administration into uniquely vulnerable tissues, such as the central nervous system or the intrathecal/epidural space, generally represent the highest level of toxicological concerns for leachables because these tissues have limited or no ability to recover from toxicity.⁴²

⁴² FDA encourages manufacturers to communicate with the Agency regarding leachables studies and the design of toxicology study protocols for intracerebral, intrathecal, epidural, and other high-risk routes.

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The risk of interactions between packaging components and the drug product is generally lower for powder for injection than it is for liquid-based products (e.g., solutions) because interaction between the powder and the primary CCS is minimal. However, some risk factors may enhance the potential for interactions between the product and the primary CCS and lead to elevated levels of leachables. For example, an elastomeric stopper may interact with the organic solvent vapor during the lyophilization process, or it may interact with the organic solvent present in the reconstituted solution.

4. Transdermal and Transmucosal Delivery Systems

Transdermal delivery systems are combination products that deliver drug across the skin and into systemic circulation.⁴³ They present a high risk with respect to the ROA because of the systemic exposure achieved through this route. These products typically have prolonged contact with the skin. Transdermal delivery systems also frequently contain multiple components, including excipients such as penetration enhancers, solubilizers, or plasticizers/softeners, as well as adhesives or tackifiers. These components are often composed of nonpolar or organic compounds (e.g., alcohols, dimethyl sulfoxide), which increase the risk of leaching of potentially harmful substances from the CCS and increase the likelihood of interaction between the packaging components and the drug product. Excipients and components such as penetration enhancers, or mechanical features such as microneedles, are typically used to increase systemic drug absorption, but they could also facilitate skin absorption and penetration of leachables from the CCS as well as other impurities (e.g., residual monomers, solvents, degradants) from the drug product.

Transmucosal delivery systems are designed to result in significant absorption of the drug across a mucous membrane (e.g., nasal, buccal, sublingual, vaginal, rectal), which can result in avoidance or significant reduction of first-pass hepatic metabolism. Some transmucosal routes may result in increased risk of toxicity or irritation to local tissues (e.g., reproductive tissues for vaginal delivery systems, rectal tissue). Each product may present different risks based on the tissues exposed and the specific formulation and ROA.

5. Topical Dermal Drug Products

Topical dermal drug products are administered to the outer surface of the body (e.g., skin). These products are dispensed from the same CCS repeatedly (often by the end user), and they can include a variety of CCSs that facilitate drug application (e.g., rigid bottle or jar, collapsible tube, flexible pouch).

Topical dermal solutions and suspensions include products that are liquid or semisolid in nature. Liquid-based products typically have a fluid or semisolid consistency and are marketed in a single- or multiple-unit container (e.g., a rigid bottle or jar, a collapsible tube, a flexible pouch). Packaging components like a dropper or spray device may be combined into a closure at the time of initial packaging. There is an increased risk for interactions between the topical dermal

⁴³ See the draft guidance for industry *Transdermal and Topical Delivery Systems — Product Development and Quality Considerations* (November 2019). When final, this guidance will represent FDA's current thinking on this topic.

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solutions and suspensions and the CCS because they may remain in contact with the drug product solution during storage and after the product is initially opened. Additional factors to consider during the risk assessment of such products include how those products are opened, reuse of the same closure system, and the accuracy and repeatability of dose disbursement from the dispensing orifice.

Risk assessments of topical dermal drug products should account for both systematic and local effects. Topical dermal solutions and suspensions have medium-to-high potential to interact with their packaging materials because nonpolar solvents or organic compounds (e.g., alcohol) are usually added to the formulation to enhance penetration into the skin. Several factors determine the degree of penetration for topical dermal products. For example, if a low dose is applied (e.g., through a low concentration, low body surface area, or low frequency of application), the drug product does not include dermal penetration enhancers, and the drug product is applied to intact skin, there could be limited systemic exposure to potential leachables. Topical dermal solutions and suspensions that contact sensitive mucosal membranes or skin that has been broken or compromised have increased safety concerns.

6. Ophthalmic and Otic Products

Ophthalmic and otic products are intended for administration into the eye and ear, respectively; they present a high risk associated with ROA based on the vulnerability of the tissues to which the drug product is applied. Local tissue toxicity assessments are essential, with toxicity determination based partly on the concentration of leachables in the solution applied to uniquely vulnerable tissues. Ophthalmic and otic products are typically solutions and suspensions designed to minimize local tissue irritation. Although ophthalmic and otic drug product solutions may be less likely to contain solvents or have pHs that are known to enhance the release of leachables, the CCSs are often manufactured with materials, such as low-density polyethylene, that are soft to facilitate squeezing the product out of the container for application. These soft and semipermeable materials are vulnerable to potential leachables and ingress of volatile chemicals. Further, many of these CCSs need to be sterilized, and that process could contribute to an increase in leachables from the CCS.⁴⁴

7. Oral Drug Products

Depending on the packaging components, the risk of interactions of drug products with the CCS can vary for oral drug products. The risk of interactions between packaging components and a solid oral dosage form like powders, capsules, or tablets is generally considered low. Appropriately referencing the indirect food additive regulations⁴⁵ and demonstrating compliance with relevant USP standards within an application or, for products not requiring an application, within the relevant documentation at the manufacturing site, may be sufficient to establish safety and compatibility of the MOC for packaging components, provided any limitations specified in the regulations are taken into consideration.⁴⁶ However, if powders are reconstituted in their final

⁴⁴ See also the guidance for industry *Certain Ophthalmic Products: Policy Regarding Compliance With 21 CFR Part 4* (March 2022).

⁴⁵ See 21 CFR parts 174–186.

⁴⁶ See MAPP 5015.5 Rev. 1, *CMC Reviews of Type III DMFs for Packaging Materials*.

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containers in the presence of organic solvents, they are at increased risk for interactions between the packaging components and the reconstituted solution or suspension. In addition, the CCS can interact with solid oral drug products containing hygroscopic excipients such as polyethylene oxides or with the glycerin present in softgel capsules, affecting the stability of both the packaging components and the drug product.

Compared to solid oral products, oral solutions and suspensions are at higher risk of interactions between the packaging components and the drug product. As with solid oral dosage forms, referencing the indirect food additive regulations⁴⁷ and demonstrating compliance with relevant USP standards may be acceptable for oral solutions and suspensions if those regulations' limitations and use conditions are met. However, if the risk assessment identifies additional safety or compatibility concerns, such as products containing cosolvents or organic compounds that are expected to extract greater amounts of substances from plastic packaging components than the food products using the same packaging materials per 21 CFR parts 174–186, extractables/leachables assessments and a safety evaluation should be performed.⁴⁸

V. QUALITY ASSESSMENT AND CONTROL

For finished pharmaceuticals, the requirements for drug product containers and closures described in 21 CFR part 211 subpart E must be met to ensure that CCS and packaging components are suitable and safe for their intended use. For drug substances, an appropriate qualification and quality control program should be implemented as described in the ICH guidance for industry *Q7 Good Manufacturing Practice Guidance for Active Pharmaceutical Ingredients* (September 2016). The overall quality evaluation of a CCS, including a variety of tests and assessments, should employ appropriate risk-based approaches. Procedures regarding the receipt, identification, storage, handling, sampling, testing (e.g., methods and analytical reference standards, acceptance criteria or specifications, and validation information for studies and evaluations), and approval or rejection of components and drug product containers and closures must be written and followed for a drug product's CCS.⁴⁹ For containers of drug substances, FDA recommends the analytical test methods and criteria for accepting and releasing each packaging component also be written and followed. FDA's Emerging Technology Program can help address concerns or challenges related to acceptance criteria for novel container closure or packaging components.⁵⁰

⁴⁷ See 21 CFR parts 174–186.

⁴⁸ We note that ICH has a new guideline currently under development, *Q3E Impurity: Assessment and Control of Extractables and Leachables for Pharmaceuticals and Biologics*, that will address the evaluation and characterization of extractables and leachables, as well as associated toxicological risk assessments.

⁴⁹ See 21 CFR 211.80. See also, 21 CFR 211.84, and 211.94(d), for additional requirements on the testing of containers.

⁵⁰ For more information, see <https://www.fda.gov/about-fda/center-drug-evaluation-and-research-cder/emerging-technology-program>. See the guidance for industry *Advancement of Emerging Technology Applications for Pharmaceutical Innovation and Modernization* (September 2017) for information on meeting requests.

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A. Typical Quality Assessments and Control of a CCS

Tests (including methods and acceptance criteria) described in the USP are typically considered minimum standards for establishing properties and characteristics of MOC or packaging components. Use of non-USP tests may be acceptable with justification. Data supporting use of a non-USP test should include a demonstration of the test's suitability for its intended use and method validation.^{51,52} If a related USP test is available and a non-USP method is used instead, supporting data should demonstrate the equivalency or superiority of the non-USP method to the USP method.

Tests and evaluation methods for CCS quality assessments and controls should be selected based on the identified risks and potential methods of mitigating these risks. Table 2 provides an overview of the typical quality assessments for qualification and quality control of CCSs for different classes of drug products, including packaging components. The following sections describe these assessments in greater detail.

Table 2. Typical Quality Assessments of a CCS⁵³

	Oral drug products		Injectable drug products		Inhalation drug products			Topical/transdermal/transmucosal products		
Dosage forms	Oral solutions and suspensions	Oral powders, tablets, capsules	Solutions and suspensions	Powders for reconstitution (lyophilized cakes)	Inhalation solutions	Inhalation aerosols (MDIs, DPIs) and inhalation sprays	Inhalation powders	Ophthalmic solutions and suspensions, otic solutions and suspensions	Transdermal drug delivery systems	Creams, ointments, lotions, Topical dermal products (powders, topical solutions, suspensions) and topical and lingual aerosols
Characteristics										
Description	For each packaging component: - Name, product code (if available), manufacturer, physical description - Identity and specifications of MOC - Description of any additional treatments (e.g., procedures for sterilizing or depyrogenating packaging components)									
Suitability										
Protection (for each component and/or the CCS, as appropriate)	Light exposure	Light exposure	Light exposure	Light exposure	Light exposure	Light exposure	Light exposure	Light exposure	Light exposure	Light exposure
	Reactive gas	Reactive gas	Reactive gas	Reactive gas	Reactive gas	Reactive gas	Reactive gas	Reactive gas	Reactive gas	Reactive gas
	Moisture permeation	Moisture permeation	Moisture permeation	Moisture permeation	Moisture permeation	Moisture permeation	Moisture permeation	Moisture permeation	Moisture permeation	Moisture permeation
	Container closure integrity	Container closure integrity	Container closure integrity	Container closure integrity	Container closure integrity	Container closure integrity	Container closure integrity	Container closure integrity	Container closure integrity	Container closure integrity
	Solvent loss or leakage (except for powders)	Solvent loss or leakage	Solvent loss or leakage	Solvent loss or leakage	Solvent loss or leakage	Solvent loss or leakage	Solvent loss or leakage	Solvent loss or leakage	Solvent loss or leakage	Solvent loss or leakage

⁵¹ See the ICH guidance for industry *Q2(R2) Validation of Analytical Procedures: Text and Methodology* (March 2024).

⁵² See USP general notice 6.30 for use of alternative tests.

⁵³ This table does not address all quality considerations relevant to the assessment of device constituent parts of CCSs. See sections III.C and V.B.3 for related information.

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	Oral drug products		Injectable drug products		Inhalation drug products			Topical/transdermal/transmucosal products			
Dosage forms	Oral solutions and suspensions	Oral powders, tablets, capsules	Solutions and suspensions	Powders for reconstitution (lyophilized cakes)	Inhalation solutions	Inhalation aerosols (MDIs, DPIs) and inhalation sprays	Inhalation powders	Ophthalmic solutions and suspensions, otc solutions and suspensions	Transdermal drug delivery systems	Creams, ointments, lotions,	Topical dermal products (powders, topical solutions, suspensions) and topical and lingual aerosols
Characteristics	For sterile products: Container closure integrity testing (CCIT), see section V.A.2.a.i and Table 3 for further information.										
Compatibility and safety (for each component and/or the CCS, as appropriate)	Chemical composition Physicochemical tests (e.g., for elastomer, plastic, or glass) Appropriate reference to indirect food additive regulations (see section IV.B.7) Extractables and leachables Impact on product quality	Chemical composition Physicochemical tests (e.g., for elastomer, plastic, or glass) Extractables and leachables Impact on product quality Biocompatibility (biological reactivity) Particulate matter	Chemical composition Physicochemical tests (e.g., for elastomer, plastic, or glass) Extractables and leachables Impact on product quality Biocompatibility (biological reactivity) Particulate matter	Chemical composition Physicochemical tests (e.g., for elastomer, plastic, or glass) Extractables and leachables Impact on product quality Biocompatibility (biological reactivity) Particulate matter	Chemical composition Physicochemical tests (e.g., for elastomer, plastic, or glass) Extractables and leachables Impact on product quality Biocompatibility (biological reactivity) Particulate matter	Physicochemical tests (e.g., for elastomer, plastic, or glass) Extractables and leachables Impact on product quality Biocompatibility (biological reactivity) Particulate matter Coating integrity test (e.g., for coatings on metal tubes)	Physicochemical tests (e.g., for elastomer, plastic, or glass) Extractables and leachables Impact on product quality Biocompatibility (biological reactivity) Particulate matter Coating integrity test (e.g., for coatings on metal tubes)	Physicochemical tests (e.g., for elastomer, plastic, or glass) Extractables and leachables Impact on product quality Biocompatibility (biological reactivity) Particulate matter Coating integrity test (e.g., for coatings on metal tubes)	Physicochemical tests (e.g., for elastomer, plastic, or glass) Extractables and leachables Impact on product quality Biocompatibility (biological reactivity) Particulate matter Coating integrity test (e.g., for coatings on metal tubes)	Physicochemical tests (e.g., for elastomer, plastic, or glass) Extractables and leachables Impact on product quality Biocompatibility (biological reactivity) Particulate matter Coating integrity test (e.g., for coatings on metal tubes)	Physicochemical tests (e.g., for elastomer, plastic, or glass) Extractables and leachables Impact on product quality Biocompatibility (biological reactivity) Particulate matter Coating integrity test (e.g., for coatings on metal tubes)
Performance	For the assembled CCS: - Functionality tests - Drug delivery assessments, as appropriate										
Quality control	For packaging components received by the manufacturer (at release of packaging components during manufacturing): - Tests and acceptance criteria (e.g., appearance, identification of material, microbiological and physiochemical tests) - Dimensional (drawing) and performance (e.g., functionality) criteria - Methods to monitor consistency in composition, as appropriate										
Stability	Considerations of CCS evaluation as part of stability testing: - Drug product stored in final CCS is tested. - An evaluation of the CCS should be included in the stability protocol, as appropriate. - CCIT is recommended in lieu of sterility testing for sterile products. - Postapproval changes of CCS may require stability studies, as applicable.										
Additional tests	For the assembled CCS, if applicable: - Stress test (e.g., shipping validation or qualification) - Freeze-thaw and thermal cycling tests										

1. Description

The general description of each packaging component of a CCS should include the following information:

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- The product name, name and address of the manufacturer, and a physical description of the packaging component (e.g., type, MOC, dimensions and drawing, shape, color).
- The MOC (e.g., plastics, paper, metal, glass, elastomers, coatings, adhesives) identified by a specific product designation and source (name of the manufacturer).⁵⁴ Alternate MOC should also be identified. Postconsumer recycled plastic should not be used in the manufacture of a primary packaging component. If it is used for a secondary packaging component, then the safety and compatibility of the material for its intended use should be addressed appropriately.
- Descriptions of any processes or preparations that are performed on a packaging component (e.g., washing, coating, sterilization, depyrogenation).⁵⁵

2. Suitability Evaluation

The suitability evaluation is, collectively, the tests and studies used and accepted to qualify a CCS or packaging components for their intended use. Every proposed CCS for a drug product must meet certain requirements, including that it provide adequate protection against foreseeable external factors in storage and use, be composed of materials that do not alter the quality of the drug, be clean, and (when appropriate given the nature of the drug) be sterilized and depyrogenated.⁵⁶ Containers for drug substances should provide adequate protection against deterioration or contamination during transportation and recommended storage; be cleaned and sanitized to ensure that they are suitable for their intended use; and not be reactive, additive, or absorptive so as to alter the quality of drug substance.⁵⁷ Generally, the suitability evaluation includes tests to demonstrate protection, compatibility, safety, and performance of packaging components and systems. If the CCS has a performance feature such as drug delivery, the evaluation should also show that the assembled CCS functions properly.

a. Protection

USP tests⁵⁸ for light exposure or transmission, moisture permeation, and container closure integrity are generally considered sufficient for evaluating a CCS's ability to protect the drug. Testing for properties other than those described in the USP (e.g., gas transmission, solvent loss, leakage) should also be performed when appropriate. For all sterile products, container closure integrity testing (CCIT) and related data should demonstrate that the proposed CCS maintains product sterility.

⁵⁴ When possible, this information should be included in the application. Alternatively, trade secret, confidential data or information not known to the applicant may be provided by referencing a drug master file (see section VI.C).

⁵⁵ For further information, see the guidance for industry *Submission of Documentation for Sterilization Process Validation in Applications for Human and Veterinary Drug Products* (November 1994).

⁵⁶ See 21 CFR 211.84, and 211.94.

⁵⁷ See the ICH guidance for industry *Q7 Good Manufacturing Practice Guidance for Active Pharmaceutical Ingredients* (September 2016).

⁵⁸ See, e.g., USP General Chapter <659> *Packaging and Storage Requirements*.

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i. Recommendations for CCIT

CCIT is used to evaluate CCS integrity to ensure the product is protected from ingress of environmental factors. For sterile products, CCIT is used:

- To demonstrate that the CCS selected is suitable to provide an adequate sterile barrier
- To validate related manufacturing parameters (e.g., capping parameters) for the selected CCS and as an in-process test for appropriate CCS closure
- To demonstrate that the CCS maintains its integrity at the labeled storage conditions over expiry ensuring protection of the product quality and sterility, if applicable

The selected method for CCIT should include method validation for the specific CCSs, sensitivity information, and negative and positive controls. Factors to consider when selecting a method for evaluating the container's integrity include the suitability of the method for the type of CCS, the intended use of the CCS, the sensitivity of the test method, and the storage conditions of the drug product. Worst-case conditions should be considered and represented when selecting an appropriate CCIT method. In some instances, the CCS's seal integrity can be transiently compromised at extreme storage conditions or during shipping and handling, which could affect product quality and safety. For example, rubber stoppers can lose their elastic properties at extreme low temperatures, and this transient loss in seal integrity can lead to overpressure in the container caused by accumulation of ingressed gas during storage. The ingressed gas can then become trapped when the elastomers reseal after being removed from the extreme temperature. This transient loss of integrity would not be captured when the CCIT is conducted after the container is removed from the extreme conditions. Therefore, a test method that can be performed at extreme temperature conditions (e.g., freezing) should be considered.

Table 3 lists examples of commonly used CCIT methods organized by specific CCS type. The list does not represent the only tests that can be performed to demonstrate container integrity for these specific CCSs. FDA encourages innovative, advanced, and deterministic integrity testing technologies. While probabilistic integrity testing methods could be acceptable, these testing methods can be challenging to design, develop, validate, and implement. All CCIT methods have advantages and disadvantages that should be considered prior to test selection.⁵⁹

Table 3. Examples of CCIT for Common Types of CCSs

CCS	MOC	CCIT
Vials or bottles with closure	Glass or plastic	Helium leak testing Pressure and/or vacuum decay Dye ingress Mass extraction test

⁵⁹ For a thorough discussion of the merits of various test methods for different types of CCSs, see USP General Chapter <1207> *Package Integrity Evaluation — Sterile Products* and its subchapters.

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CCS	MOC	CCIT
Ampoules	Glass	Helium leak testing Pressure and/or vacuum decay Dye ingress High voltage leak detection
	Plastic blow fill seal ^a	Helium leak testing Pressure and/or vacuum decay Dye ingress High voltage leak detection
Pre-filled syringes/ cartridges	Glass or plastic	Helium leak testing Pressure and/or vacuum decay Dye ingress Mass extraction test
Tubes	Plastic or Aluminum	Dye ingress
Flexible bags	Plastic	Helium leak testing Pressure and/or vacuum decay Dye ingress Mass extraction test Internal pressurization (bubble emission test)
Blister	Plastic, aluminum foil	Helium leak testing Dye ingress Seal integrity testing Burst/creep testing
Sachets/pouches	Foil/paper	Pressure and/or vacuum decay
	Plastic	Pressure and/or vacuum decay Dye ingress

^a Container closure defects can be a major problem for blow-fill-seal operations. A reliable, sensitive leak detection test should be used to inspect each unit of each batch. Several leak detection methods, including high voltage leak detection or vacuum leak detection, can be adapted easily for in-line leak detection to ensure product container integrity. For more information, see the guidance for industry *Sterile Drug Products Produced by Aseptic Processing — Current Good Manufacturing Practice* (September 2004).

All sterile products must be tested to verify the maintenance of sterility over the drug product's shelf life.⁶⁰ Because of the limitations of sterility testing, the Agency encourages performing CCIT in lieu of sterility testing as a component of a stability program to ensure that containers are able to maintain sterility throughout the drug product's shelf life.⁶¹

b. Safety and compatibility

To ensure that the CCS is safe and compatible with the drug, packaging components and MOC should be well characterized. Characterization should include the complete chemical composition of every material used in the manufacture of each packaging component and should

⁶⁰ See 21 CFR 211.166, 21 CFR 211.167(a), and 21 CFR 610.12.

⁶¹ See the guidance for industry *Container and Closure System Integrity Testing in Lieu of Sterility Testing as a Component of the Stability Protocol for Sterile Products* (February 2008).

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include results of appropriate qualification and characterization tests (e.g., USP tests for elastomers, plastic, or glass).⁶² For application products, this information may be provided as a reference to Type III master files.

For drugs that are unlikely to interact with the CCS, other tests or information (e.g., appropriate reference to the indirect food additive regulations at 21 CFR parts 174–186) can be used to address the issue of safety and compatibility. For example, safety and compatibility of the CCS for an oral solid could be addressed by meeting compendial requirements (e.g., USP tests for glass and plastic containers) and conforming to the requirements for indirect food additives provided in the CFR (e.g., 21 CFR 177.1520 for olefin polymers).

i. Extractables and leachables assessments

Extractables and leachables assessment is an integral part of a CCS's suitability evaluation.⁶³ These studies and associated toxicological assessments should be performed for any drug that is likely to interact with its packaging components, leading to introduction of leached substances into the patient. A risk-based approach should be used when designing and assessing extractables and leachables studies for a CCS and when making changes to packaging components during the product's life cycle. Risk assessment for leachables should be performed on a case-by-case basis (i.e., it should be product-specific).

In general, leachables studies are performed throughout the shelf life of a drug product. These studies are conducted under specified storage conditions as part of formal stability studies in which multiple batches (e.g., three batches) of the drug product are tested at multiple time points under both accelerated and long-term stability conditions, as well as intermediate conditions when appropriate.⁶⁴ A justification should be provided for selections of batches other than the formal stability batches or if fewer than three batches are selected. The CCS (including secondary components such as labels, adhesives, over-pouches, wraps, and cartons) used in leachables studies should be representative of the future commercial product. The analytical methods used for leachables studies are typically the same as, or similar to, those used for the extractables studies and should be properly validated. A correlation between extractables and leachables could be established when sufficient extractables and leachables data from the same CCS are generated.

⁶² USP testing is described in USP General Chapters <381> *Elastomeric Components in Injectable Pharmaceutical Product Packaging/Delivery Systems*, <660> *Containers — Glass*, and <661> *Plastic Packaging Systems and Their Materials of Construction*. Note that USP permits early adoption of the requirements in USP General Chapters <661>, <661.1> *Plastic Materials of Construction*, and <661.2> *Plastic Packaging Systems for Pharmaceutical Use*.

⁶³ We note that ICH has a new guideline currently under development, *Q3E Impurity: Assessment and Control of Extractables and Leachables for Pharmaceuticals and Biologics*, that will address the evaluation and characterization of extractables and leachables, as well as associated toxicological risk assessments. Also see USP General Chapters <1663> *Assessment of Extractables Associated with Pharmaceutical Packaging/Delivery Systems*, <1664> *Assessment of Drug Product Leachables Associated with Pharmaceutical Packaging/Delivery Systems*, and others also describe extractables and leachables assessments.

⁶⁴ For general guidance on conducting stability studies, see the ICH guidances for industry *Q1A(R2) Stability Testing of New Drug Substances and Products* (November 2003) and *Q5C Quality of Biotechnological Products: Stability Testing of Biotechnological/Biological Products* (July 1996).

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For drug substances, as appropriate, extractables and leachables studies and associated toxicological assessments should support the evaluation of leachates from the container closure components.

ii. Nonclinical toxicological risk assessments

A toxicological risk assessment of the leachables from a CCS should be provided to support the CCS's safety for drugs.⁶⁵ A toxicological risk assessment addressing general toxicological concerns should be provided for all leachables exceeding the recommended **qualification threshold (QT)**. A toxicological risk assessment should also be provided for any leachables with identified concerns for DNA reactivity or carcinogenicity if they exceed the appropriate threshold of toxicologic concern-based acceptable intake (TTC-based AI). When local tissue toxicity is a primary concern, a concentration-based QT may also be appropriate. The **safety concern threshold (SCT)** should be based on the TTC-based AI or the QT, whichever is lowest. The **analytical evaluation threshold (AET)** should be based on the SCT established for a given CCS for a drug. FDA generally recommends the following:

- An SCT of 1.5 mcg/day should be applied for most chronic-use drugs.
- A QT should be applied for general toxicity for all products.
- A QT based on concentration also applies for drugs in which local tissue toxicity is of concern (e.g., ophthalmic, intrathecal, perineural).
- A lower SCT may be recommended if the CCS may contain compounds of specific toxicological concern (e.g., polynuclear aromatic hydrocarbons, nitrosamines).

The toxicological risk assessment should be based on the highest level of confirmed leachables present in the product over the proposed shelf life. Leachables exceeding the AET should be identified in a leachables study that is based on the results of extraction studies performed with appropriate extraction solvents. The leachables should be identified and quantified using validated analytical methods.

c. Performance

Tests and acceptance criteria used to evaluate CCS performance should be appropriate for the drug's particular dosage form, ROA, and CCS design features.

To address performance, functionality tests⁶⁶ are considered sufficient if the test and acceptance criteria are appropriate for the CCS's intended purpose. Performance evaluations should include an analysis of how storage conditions and potential stress from material physical form transformation (such as crystallization during a lyophilization process) could affect the CCS's

⁶⁵ We note that ICH has a new guideline currently under development, *Q3E Impurity: Assessment and Control of Extractables and Leachables for Pharmaceuticals and Biologics*, that will address the evaluation and characterization of extractables and leachables, as well as associated toxicological risk assessments.

⁶⁶ Recommendations for functionality tests are described in USP General Chapters, including <659> *Packaging and Storage Requirements*, <381> *Elastomeric Components in Injectable Pharmaceutical Product Packaging/Delivery Systems*, and <671> *Containers — Performance Testing*.

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functionality. A stability program should include a performance evaluation at the end of a product's shelf life to ensure the CCS maintains functionality as designed through expiry. As described in section III.C, the CCS can serve in a dual capacity: it can contain and protect the drug, and it can also act to deliver the drug as a device constituent part of a combination product. The type of performance testing will depend on the type of CCS and its intended function as a device constituent part (see also section V.B.3). Testing with a stand-alone device may also be needed if the CCS is labeled for use with a device(s) (e.g., a drug cartridge labeled for use with a separate, reusable pen injector) and the CCS performance can impact overall device and drug product performance.

3. Quality Control

Quality control is the use of tests and acceptance criteria to ensure that packaging components and the CCS continue to possess the characteristics established in the suitability studies.^{67,68} A manufacturer's quality system should ensure that all container and closure suppliers are reliable and that quality controls have been established for receipt, production, storage, and use. A robust quality system should have a supplier qualification program that accounts for variations in component specifications from different suppliers. Sufficient initial tests should be performed to establish reliability, and subsequent tests should be performed periodically to verify reliability.⁶⁹ Manufacturers should take into account that a change to a packaging component may result in a new or revised specification for the packaging component.

Dimensional information for the CCS components (including device constituent parts) typically contains a detailed schematic drawing complete with target dimensions and tolerances, including test methods and acceptance criteria for each CCS component. A separate drawing may not be necessary for a CCS component (including device constituent parts) if it is part of a larger unit in a drawing or if the component is uncomplicated in design (e.g., a cap liner). Dimension information should be provided in an application, or maintained at the manufacturing site, as appropriate.

4. Stability Testing

Stability testing of a drug should be conducted in the CCS proposed for marketing (including, as appropriate, any secondary packaging).⁷⁰ When applicable, the stability testing protocol should include evaluation of additional functionality of the CCS (e.g., dose uniformity of an MDI until

⁶⁷ See 21 CFR 211.84 and 211.94.

⁶⁸ A CCS that is also a device constituent part is subject to 21 CFR part 820 requirements in accordance with 21 CFR part 4, subpart A, including design controls (see section V.B.3) and purchasing controls, among other requirements. The Agency has exempted some devices from all or certain provisions of the device quality system regulations. Such exemptions may extend to the device constituent parts of combination products and to the combination products of which they are a part. See the guidance for industry and FDA staff *Current Good Manufacturing Practice Requirements for Combination Products* (January 2017) for more information.

⁶⁹ See the ICH guidance for industry *Q10 Pharmaceutical Quality System* (April 2009).

⁷⁰ For general guidance on conducting stability studies, see the ICH guidances for industry *Q1A(R2) Stability Testing of New Drug Substances and Products* (November 2003) and *Q5C Quality of Biotechnological Products: Stability Testing of Biotechnological/Biological Products* (July 1996). See also the guidances for industry *ANDAs: Stability Testing of Drug Substances and Products* (June 2013) and *ANDAs: Stability Testing of Drug Substances and Products, Questions and Answers* (May 2014).

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expiry). Even when a formal test for quality of the CCS is not performed, the manufacturer should investigate any observed change in the CCS during the stability studies. For sterile products, maintenance of sterility throughout the product's shelf life can typically be demonstrated by container closure integrity testing conducted annually, and at expiry, in lieu of sterility testing (see section V.A.2.a.i for more information about CCIT).

5. Additional Tests for Shipping, Handling, and Storage

A shipping study may be performed by evaluating the integrity of the CCS after the drug has been shipped from its manufacturing site to its intended destination (e.g., a packaging/labeling, storage, or distribution center) using the intended commercial shipping method and under the recommended storage conditions.⁷¹ The shipping study of the drug may include evaluating the mechanical protective function of the CCS, which should not have been compromised or otherwise affected during shipping. An integrity evaluation of a CCS should also be considered after shipping sterile drugs.

Stress testing used in a shipping study ensures that a CCS adequately protects the drug from both environmental stress (e.g., light, reactive gas permeation, moisture permeation) and physical stress (e.g., compression, shock, vibration, change in air pressure, expansion) during transportation and handling. The design may include packaging components that supplement the CCS's mechanical strength to enhance its protective function.

Stress testing may include laboratory simulation studies using appropriate test protocols⁷² to evaluate the CCS's protective function. The CCS should be able to withstand simulated mechanical stress in order to demonstrate it provides adequate protection for the drug and the accompanying labels during shipping and handling. Both destructive and nondestructive evaluation techniques could be used to measure the integrity of the CCS (see section V.A.2.a.i and Table 3). Destructive evaluation techniques (e.g., opening the opaque blister pack) are typically used to evaluate the physical integrity of a CCS (e.g., fragile ampules, vials, bottles, device constituent parts).

For drugs shipped under cold-chain conditions (e.g., frozen or refrigerated), the shipping containers should be appropriately qualified using temperature recording systems to ensure the intended product temperature is maintained when the CCS is exposed to the worst-case temperature and shipping time expected. The qualification of the CCS under specified temperature conditions (e.g., insulated dry ice box or temperature-controlled shipping container) is typically performed using temperature recording instruments as part of shipping validation.

When applicable, drug development should include freeze-thaw and thermal cycling studies to demonstrate that drug quality and CCS integrity and functionality can be maintained under temperatures that represent both normal and worst-case transportation, storage, and handling

⁷¹ For drugs that are sensitive to environmental stresses, the evaluation of critical quality attributes of drugs prior to or after shipping and handling should be performed to confirm the protective function of the CCS, as applicable.

⁷² Refer to the American Society for Testing and Materials (ASTM)'s standards as applicable, and the International Safe Transit Association (ISTA)'s *Guidelines for Selecting and Using ISTA Test Procedures and Projects*.

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conditions.⁷³ These studies typically include exposing the drug, while in the intended commercial CCS, to multiple freeze-thaw cycles or thermal cycles. The assessment of CCS integrity and functionality may include the visual inspection of defects (e.g., cracks, breakage, particulates) and CCIT as recommended in Table 3.

B. Special Considerations for Particular Packaging Components

While all primary packaging components of CCSs should undergo quality assessment, some CCSs have additional packaging components with special considerations for quality assessment. This section describes the tests and studies recommended to establish suitability and quality control for secondary packaging components, auxiliary components, and packaging components that are device constituent parts of a combination product.

1. Secondary Packaging Components

Secondary packaging components do not make contact with the drug, so there is less concern that the drug will interact with the secondary package's MOC (e.g., cartons made of paper or plastic; overwraps made of metal foil, plastic, or paper). Secondary packaging can have an additional protective function, as when it prevents external factors (e.g., moisture, light, microbial contaminants) from reaching the primary CCS, or when it provides added protection for a primary CCS that is flexible or subject to rough handling. For example, the CCS for inhalation aerosols and inhalation powders may include overwraps or foil pouches as secondary packaging in addition to the inhaler. Secondary packaging components are also used to maintain surface sterility of a primary CCS intended for use in a sterile surgical field.

The amount of information needed to demonstrate a secondary packaging's suitability depends on the packaging's purpose. Typically, a brief description of a secondary packaging component is sufficient. If the secondary packaging component is intended to provide additional protection or function beyond container closure (e.g., drug delivery, controlled access to the drug, electronic monitoring of the drug), appropriate assessments should demonstrate that the component provides the additional protection or functions as intended, without adversely interacting with the drug. In such cases, relevant in-process controls should be established with additional testing, including seal integrity testing, stability studies of the drug within the secondary packaging, and in-use stability testing of the drug once the protective secondary packaging is opened, as applicable.

If the primary CCS is relatively permeable, the drug could be contaminated by the migration of an ink, an adhesive component, or a volatile substance present in the secondary packaging

⁷³ See the ICH guidances for industry *Q1A(R2) Stability Testing of New Drug Substances and Products* (November 2003); *Q5C Quality of Biotechnological Products: Stability Testing of Biotechnological/Biological Products* (July 1996); and World Health Organization, 2018, *Stability Testing of Active Pharmaceutical Ingredients and Finished Pharmaceutical Products*, In: WHO Expert Committee on Specifications for Pharmaceutical Preparations: Fifty-Second Report, Annex 10 (WHO Technical Report Series, No. 1010). Also see USP General Chapter <1079> *Good Storage and Shipping Practices*. See also Parenteral Drug Association, 2005, Technical Report #39: Cold Chain Guidance for Medicinal Products: Maintaining the Quality of Temperature-Sensitive Medicinal Products Through the Transportation Environment, PDA J Pharm Sci Technol, 59(3 Suppl TR39):1–12.

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component. If the secondary packaging component is a potential source of contamination, the safety of its MOC should be demonstrated.

2. Auxiliary Packaging Components

Auxiliary packaging components are articles that are used to support or enhance a CCS. These articles include, but are not limited to, pharmaceutical coil (e.g., cotton, rayon), desiccants, and oxygen scavengers. Pharmaceutical coil, desiccants, and oxygen scavengers included in drug primary packaging prevent tablet and capsule breakage and exposure to moisture and oxygen, respectively. These types of packaging components should be chosen carefully to prevent unintended ingestion of desiccants by the consumer or unacceptable toxicity from any materials used. The auxiliary component (e.g., desiccants) should differ in shape and/or size from the dosage form (e.g., tablets, capsules) with which it is packaged. The quality standards described in USP General Chapter <670> *Auxiliary Packaging Components* are typically considered sufficient to establish the suitability of pharmaceutical coil and desiccants. If other auxiliary packaging components are used, the composition and appropriate tests with acceptance criteria to establish suitability should be provided (see section V.A.2 for further information about suitability evaluations). Stability studies must use the final CCS in which the drug product will be marketed, including any auxiliary packaging component.⁷⁴ Stability studies of the drug substance should also be conducted in a CCS that is the same as or simulates the packaging proposed for storage and distribution, including any auxiliary packaging components.⁷⁵

3. Packaging Components for Device Constituent Parts of Combination Products

A CCS that is also a device constituent part of a combination product is subject to certain regulatory requirements that are beyond the scope of this guidance.⁷⁶ These requirements include, but are not limited to, CGMP⁷⁷ and postmarketing safety reporting⁷⁸ requirements under 21 CFR part 4 and the submission of information in premarket applications to support the safety and effectiveness of the combination product, including the performance of the device constituent part.^{79,80,81} For example, if the CCS is a device constituent part of a combination product, the device constituent part and combination product as a whole must be designed and

⁷⁴ See 21 CFR 211.166(a)(4).

⁷⁵ See the ICH guidance for industry *Q1A(R2) Stability Testing of New Drug Substances and Products* (November 2003).

⁷⁶ Applicants who have questions regarding combination product requirements should contact the lead center or the Office of Combination Products (combination@fda.hhs.gov) for assistance.

⁷⁷ For further information, see the guidance for industry and FDA staff *Current Good Manufacturing Practice Requirements for Combination Products* (January 2017).

⁷⁸ See the guidance for industry and FDA staff *Postmarketing Safety Reporting for Combination Products* (July 2019).

⁷⁹ For further information, see the guidance for industry and FDA staff *Principles of Premarket Pathways for Combination Products* (January 2022).

⁸⁰ FDA's approval of an abbreviated new drug application is based on demonstration of sameness to the applicable reference listed drug, which then allows FDA to rely on its previous findings of the safety and effectiveness of the reference listed drug. See section 505(j) of the FD&C Act.

⁸¹ FDA's clearance of a 510(k) is based on a demonstration of substantial equivalence (which is basically a comparative safety and effectiveness evaluation) to a legally marketed predicate device. See section 513(i) of the FD&C Act.

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tested consistent with the requirements in Clause 7.3 of ISO 13485:2016, as required by 21 CFR 820.10(c), if applicable.⁸² The design control requirements in Clause 7.3 of ISO 13485:2016, as required by 21 CFR 820.10(c) may overlap with the requirements and recommendations for CCS quality assessments and controls discussed in this guidance. A CCS that is a device constituent part of a combination product may also be subject to purchasing controls requirements in Clause 7.4 of ISO 13485:2016, as required by 21 CFR 820.10(a) for any container closure components that are supplied by another manufacturer. Similarly, purchasing control requirements relate to, and may overlap with, CCS quality assessments and controls such as those discussed in this guidance.

For a CCS that is also a device constituent part of a combination product, appropriate testing should be performed to demonstrate the device's function and delivery performance.⁸³ Evaluations should verify and validate the drug delivery performance after shipping, during storage, and in use.⁸⁴ For example, pre-filled syringes serve as a delivery device by action of the plunger system; the elastomeric component (i.e., the plunger) must move to empty the container upon demand. Tests of this function (e.g., glide force, break force) as well as the device's container closure function (e.g., seal integrity tests), should be provided to help evaluate these systems. The CCS for an MDI generally consists of the device constituent parts (i.e., the canister, the actuator, and the metering valve). To characterize the delivery performance of an MDI, the application should provide data to demonstrate consistent and reliable dose delivery (e.g., valve delivery, delivered dose uniformity, aerodynamic particle size distribution, spray pattern).⁸⁵ Other assessments relating to device performance may include human factors studies to evaluate the user interface of a product.⁸⁶

⁸² As explained in the guidance for industry and FDA staff *Current Good Manufacturing Practice Requirements for Combination Products* (January 2017), the Agency has exempted some devices from all or certain provisions of the device quality system regulations. For example, 21 CFR 880.6430 exempts liquid medication dispensers (cups, droppers, etc.) from all provisions of the device quality system regulations with the exception of requirements concerning records and complaint files in Clauses 4.2.5 and 8.2.2 of ISO 13485:2016, among other requirements under 21 CFR 820.35 (recordkeeping and complaint file requirements were previously in 21 CFR 820.180 and 820.198, respectively). Such exemptions may extend to device constituent parts of combination products and to the combination products of which they are a part. If the exemptions for a device constituent part of a drug-device combination product cover all of the part 820 provisions included in 21 CFR 4.4(b)(1), then the Agency will consider the combination product manufacturer CGMP compliant so long as the CGMP operating system is compliant with part 211 (no demonstration of compliance with part 820 will be necessary). However, if a device that would ordinarily be exempt from all or certain provisions in part 820 is incorporated into a CCS, for example, if a dropper is incorporated into the cap of a bottle of a drug, this may be a new use of the device such that the exemptions from part 820 may not be applicable. In any event, when incorporated into a CCS, the device must be addressed as part of the CCS for purposes of part 211. For further information, see the guidance for industry and FDA staff *Current Good Manufacturing Practice Requirements for Combination Products* (January 2017).

⁸³ See draft guidance for industry *Essential Drug Delivery Outputs for Devices Intended to Deliver Drugs and Biological Products* (June 2024). When final, this guidance will represent FDA's current thinking on this topic.

⁸⁴ Ibid.

⁸⁵ See the draft guidance for industry *Metered Dose Inhaler (MDI) and Dry Powder Inhaler (DPI) Products — Quality Considerations* (April 2018). When final, this guidance will represent FDA's current thinking on this topic.

⁸⁶ See guidance for industry and FDA staff *Application of Human Factors Engineering Principles for Combination Products: Questions and Answers* (September 2023). See also the draft guidance for industry *Comparative Analyses and Related Comparative Use Human Factors Studies for a Drug-Device Combination Product Submitted in an ANDA* (January 2017). When final, this guidance will represent FDA's current thinking on this topic. The recommendations included in this guidance generally focus on the analysis of the proposed user interface for a

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C. Recommendations for Bulk Containers

A bulk container can be used to store and ship drug substances and bulk drug products.⁸⁷ The bulk container should adequately protect the drug and should be constructed of materials that are safe and are compatible with the drug.

The quality assessments regarding the bulk container used for drug substances should generally include:

- a detailed description of the complete CCS for the bulk drug substance (e.g., specific container, closure, all liners, inner seal, desiccant)
- the composition of each component
- the analytical test methods and criteria for the acceptance and release of each packaging component

A bulk container may be used to transport bulk drug products for filling, packaging, or repackaging. The bulk container should meet the same requirements for protection, compatibility, and safety as the final or primary CCS during the specified storage time and conditions of the bulk drug products.⁸⁸

VI. CCS INFORMATION SUBMITTED IN SPECIFIC APPLICATION TYPES

For products that require an application, information about the CCS should be included directly in the application, or in a master file that may be included by reference in an application, to show that each proposed CCS and its components are suitable for their intended use. This section addresses the particular types of CCS information that should be submitted in different application types.

generic drug-device combination product (generic combination product) when compared to the user interface for a reference listed drug. For information regarding the type and amount of human factor data necessary for biosimilars and interchangeables, please contact the Division of Medication Error Prevention and Analysis within CDER's Office of Medication Error Prevention and Risk Management.

⁸⁷ A bulk container is not the final CCS for finished drug products; instead, it is an intermediate container that holds the bulk drug for further manufacturing or packaging, or for shipment to repackagers or contract packagers. Pharmacy bulk packages (PBPs) and Imaging Bulk Packages (IBP) are not considered bulk containers for purposes of this guidance because they are final containers of marketed drug products in large quantities. Designation as a PBP or IBP is limited to injection, for injection, or injectable emulsion dosage forms as defined in USP General Chapter <1121> *Nomenclature*. See USP General Chapter <659> *Packaging and Storage Requirements* for the definitions of package type terms.

⁸⁸ When the drug will be subject to a new drug application or a biologics license application (under section 351(a) or (k) of the PHS Act), FDA recommends contacting the specific drug product review division with questions about bulk containers. For further information, see the draft guidances for industry *Formal Meetings Between the FDA and Sponsors or Applicants of PDUFA Products* (Rev 1) (September 2023) and *Formal Meetings Between the FDA and Sponsors or Applicants of BsUFA Products* (Rev 1) (August 2023). When final, these guidances will represent FDA's current thinking on these topics. When the submission is an abbreviated new drug application (ANDA), submit questions through the pre-ANDA meeting or controlled correspondence processes. See the guidances for industry *Formal Meetings Between FDA and ANDA Applicants of Complex Products Under GDUFA* (Rev 1) (October 2022) and *Controlled Correspondence Related to Generic Drug Development* (Rev 1) (March 2024).

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A. Marketing Applications

All significant phases of the manufacturing and processing of a drug should be described in the CMC section of the application for marketing approval or in a master file referenced in the application.⁸⁹ Use of the electronic common technical document (eCTD) format is required when submitting the quality section of an application to FDA.⁹⁰ Other FDA guidance provides for the required formatting of device constituent part information.⁹¹

Table 4 summarizes the general information that should be submitted for a CCS in an application for marketing approval. The recommended information should also be included in supplements to report changes to approved applications.

Table 4. General Information That Should Be Submitted for a CCS in an Application for Marketing Approval

eCTD references	Recommended information
CCS for drug substances	
2.3.S.6	A brief description and discussion of the CCS information for the drug substance (see 3.2.S.6), with summary tables and hyperlinks to related testing reports
3.2.S.6	Description • Identity of MOC • Specifications Suitability • Choice of materials • Protection • Compatibility • Safety • Leachables and extractables assessments • Container closure integrity for sterile substances
3.2.S.7	Stability

⁸⁹ With a few exceptions, FDA does not generally permit biological products in biologics license applications to incorporate by reference drug substance, drug substance intermediate, or drug product information in master files (referred to as *Type II drug master files*) (see 21 CFR 601.2(g)(1)).

⁹⁰ Electronic submission and use of the eCTD format are required for investigational new drug applications, new drug applications, abbreviated new drug applications, biologics license applications, and most drug master files. See section 745A(a) of the FD&C Act; guidance for industry *Providing Regulatory Submissions in Electronic Format — Certain Human Pharmaceutical Product Applications and Related Submissions Using the eCTD Specifications* (September 2024). For further information, see the ICH guidances for industry *M4Q: The CTD — Quality* (August 2001) and *M4S: The CTD — Safety* (August 2001). See also the guidance for industry *ANDA Submissions — Content and Format of Abbreviated New Drug Applications* (June 2019).

⁹¹ For information about the device constituent part portion of the application, see the technical specifications document *eCTD Technical Conformance Guide* (November 2022) and the guidance for industry *Providing Regulatory Submissions in Electronic Format — Certain Human Pharmaceutical Product Applications and Related Submissions Using the eCTD Specifications* (February 2020).

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eCTD references	Recommended information
CCS for drug products	
2.3.P.7	A brief description and discussion of the CCS information for the drug product (see 3.2.P.7), with summary tables and hyperlinks to related testing reports
3.2.P.2.4	Suitability • Choice of materials • Protection • Compatibility • Safety • Performance^a • Leachables and extractables assessments
3.2.P.2.5	Microbiological attributes ^b • Demonstrate container closure integrity for sterile products
3.2.P.3.5	Process validation and/or evaluation for primary container closure • Cleaning • Depyrogenation • Sterilization validation
3.2.P.7	Description • Identification of MOC • Specifications and certificate of analysis • Engineering drawing • Test results per USP requirements
3.2.P.8	Stability
Toxicology of Extractables and Leachables	
2.6.6.8	A summary and discussion of toxicological risk assessments of leachables and extractables from 3.2.S.6, 3.2.P.2, and 3.2.P.7, and the rationale for conducting the studies, including summary tables and hyperlinks to related testing reports
4.2.3.7.7	Nonclinical toxicological risk assessments

^a If a CCS is also a device constituent part, the application should provide all appropriate additional information.

^b For microbiological attribute considerations of nonsterile products, see the draft guidance for industry *Microbiological Quality Considerations in Non-Sterile Drug Manufacturing* (September 2021). When final, this guidance will represent FDA’s current thinking on this topic.

B. Investigational Applications⁹²

The CMC section of an investigational new drug application typically includes brief descriptions of the packaging components, the assembled CCS, and any precautions needed to ensure that the drug is protected and preserved during its use in clinical trials. The amount and depth of CMC information that applicants should submit in the investigational new drug application depends on the phase of the investigation, the nature of the clinical study proposed in humans, and whether

⁹² “Investigational applications” as used here refers to investigational new drug applications (INDs) as well as investigational device exemption applications (IDEs) for device-led combination products. For IDEs, refer to 21 CFR 812.27 for submission requirements for nonclinical testing. The type and extent of testing appropriate for CCSs of the drug are generally the same for both IDEs and INDs.

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the information is safety-related. FDA may request that additional, more comprehensive CCS quality information be submitted to address specific safety concerns (e.g., proposals to employ new or novel packaging materials or for atypical delivery systems).⁹³ Updates to the CCS information should be provided if it differs from that reported in the studies for the previous phase. If changes that could affect product quality are made to the CCS, the applicant should submit an information amendment (e.g., CMC safety information).

C. Master Files

Drug master files (DMFs) may be used to provide confidential, detailed information about packaging of drug products. DMF holders can authorize an applicant to incorporate by reference information contained in the DMF without having to disclose that information to the applicant.^{94,95} For example, type II drug master files might be used to provide container closure information for drug substances and, if applicable, CCIT information for sterile drug substances; type III drug master files might be used to provide information about packaging materials, including descriptions of the MOC and packaging components, controls for packaging materials and components, and data (e.g., suitability evaluation) supporting acceptability of the packaging material or component for its intended use. Type III drug master files can also include information regarding depyrogenation and sterilization of packaging components, as applicable.

VII. REPORTING POSTAPPROVAL CHANGES TO A CCS

FDA's regulations describe when a change to an approved drug product (including a biological product) or its manufacturing process requires a postmarket submission. The provisions also apply to changes to a CCS component, a device constituent part that is (or is part of) the CCS, or a CCS component's MOC, and to changes to a manufacturing process involving any of these items.⁹⁶

⁹³ For general guidance regarding the CCS information to be submitted for phase 1, 2, and 3 studies of an investigational new drug application, see the guidances for industry *Content and Format of Investigational New Drug Applications (INDs) for Phase 1 Studies of Drugs, Including Well-Characterized, Therapeutic, Biotechnology-Derived Products* (November 1995) and *INDs for Phase 2 and Phase 3 Studies: Chemistry, Manufacturing, and Controls Information* (May 2003).

⁹⁴ For general information about master file format and content, see the draft guidance for industry *Drug Master Files* (October 2019). When final, this guidance will represent FDA's current thinking on this topic. For information specific to drugs, see <https://www.fda.gov/drugs/forms-submission-requirements/drug-master-files-dmfs>. For information specific to biological drugs, see <https://www.fda.gov/vaccines-blood-biologics/development-approval-process-cber/master-files-cber-regulated-products>. For information specific to devices and device constituent parts, see <https://www.fda.gov/medical-devices/premarket-approval-pma/master-files>.

⁹⁵ FDA does not approve drug master file submissions, but rather reviews the technical contents of the master file in support of the application. An IND, NDA, or ANDA that incorporates by reference information in a drug master file must include a letter of authorization from the drug master file holder to the application holder. See 21 CFR 314.420(b). Each incorporation by reference must describe the information being incorporated by name, reference number, volume, and page number.

⁹⁶ The requirements are specified in 21 CFR 314.70 for a new drug application, in 21 CFR 314.97 for an abbreviated new drug application, and in 21 CFR 601.12 for a biologics license application. All manufacturing changes must be reviewed and approved by the quality control unit (see 21 CFR 211.100(a) and 211.160(a)).

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Any changes to a drug's CCS should be based on a risk assessment for potential effects on identity, strength, quality, purity, or potency (see section IV). The type of information that should be submitted depends on the potential for adverse effects on the safety or effectiveness of the drug. In some cases, there may be substantial potential for adverse effects, even if testing was conducted to establish conformance with the approved specification. A change to a packaging component may result in a new or revised specification for the packaging component.⁹⁷

Multiple guidances for industry describe the Agency's recommendations for how postapproval changes to CCSs should be reported.⁹⁸ Submissions should include risk analyses and quality-related evaluations as described in sections IV and V of this guidance⁹⁹ and should provide a scientific justification for omitting any recommended testing. Changes to a device constituent part of a combination product affect the combination product as a whole; for example, a change in the device constituent may change the container closure function, device performance characteristics, and the performance and labeling of the overall combination product. Therefore, when considering the data to provide for a change to a device constituent part of a combination product, in addition to addressing the change in the container closure (as described in the referenced guidance documents), applicants should consult other relevant guidance documents.¹⁰⁰

⁹⁷ In accordance with the ICH guidance for industry *Q10 Pharmaceutical Quality System* (April 2009), the manufacturer's quality system should be considered to extend to control of the quality of purchased materials. A firm's supplier qualification program should take into account that component specifications may vary by supplier.

⁹⁸ Some of the guidances that may be relevant are: the ICH guidance for industry *Q12 Technical and Regulatory Considerations for Pharmaceutical Product Lifecycle Management* (May 2021) and the guidances for industry *Changes to an Approved NDA or ANDA* (April 2004), *Changes to an Approved NDA or ANDA: Questions and Answers* (January 2001), *CMC Postapproval Manufacturing Changes To Be Documented in Annual Reports* (March 2014), *CMC Postapproval Manufacturing Changes for Specified Biological Products To Be Documented in Annual Reports* (December 2021), and *Chemistry, Manufacturing, and Controls Changes to an Approved Application: Certain Biological Products* (June 2021). See also the draft guidances for industry *Postapproval Changes to Drug Substances* (September 2018), *Submissions for Postapproval Modifications to a Combination Product Approved Under a BLA, NDA, or PMA* (January 2013), and *ICH Q12: Implementation Considerations for FDA-Regulated Products* (May 2021). When final, these guidances will represent FDA's current thinking on these topics.

⁹⁹ See the ICH guidances for industry *Q9(R1) Quality Risk Management* (May 2023) and *Q10 Pharmaceutical Quality System* (April 2009).

¹⁰⁰ FDA has published the following draft guidances which may be a helpful resource: *Submissions for Postapproval Modifications to a Combination Product Approved Under a BLA, NDA, or PMA* (January 2013), *Bridging for Drug-Device and Biologic-Device Combination Products* (December 2019), and *Essential Drug Delivery Outputs for Devices Intended to Deliver Drugs and Biological Products* (June 2024). When final, these guidances will represent FDA's current thinking on these topics.

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GLOSSARY

Analytical evaluation threshold (AET): The threshold at or above which a particular extractable or leachable should be identified, quantified, and further reported for potential toxicological assessment. AET is based directly on the SCT, but it is a relative value, not an absolute value.

Combination product: A combination product is a human medical product composed of two or more different types of medical products (i.e., drug and device, drug and biological product, device and biological product, or all three together) (see 21 CFR 3.2(e)). A constituent part of a combination product is a drug, device, or biological product that is a part of the combination product (see 21 CFR 4.2).

Container closure system: The sum of packaging components and materials that together contain and protect the article. This includes primary packaging components, and it includes secondary packaging components when they are required to provide additional protection.

Critical quality attribute: A critical quality attribute is a physical, chemical, biological, or microbiological property or characteristic that should be within an appropriate limit, range, or distribution to ensure the desired product quality. (ICH Q8(R2))

Extractables: Organic and inorganic chemical entities that can be released from a pharmaceutical packaging or delivery system, packaging component, or packaging material of construction and into an extraction solvent under laboratory conditions.

Leachables: Foreign organic and inorganic chemical entities that are present in a packaged drug product because they have leached into the packaged drug product from a packaging or delivery system, packaging component, or packaging material of construction under normal conditions of storage and use or during accelerated drug product stability studies.

Materials of construction: The materials (e.g., glass, plastic, elastomers, metal) of which a packaging component consists.

Packaging component: Any single part of the package or container closure system, including the container (e.g., ampules, syringes, vials, bottles), closures (e.g., screw caps, stoppers), ferrules and overseals, closure liners (e.g., tube cartridge liners), inner seals, administration ports, overwraps, administration accessories, labels, cardboard boxes, and shrink wrap.

Primary packaging component: A packaging component that is in direct contact with, or may come into direct contact with, the drug.

Qualification threshold (QT): The threshold below which a given leachable is not considered for safety qualification unless the leachable presents structure–activity relationship concerns.

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1000 **Safety concern threshold (SCT):** The threshold below which a leachable would have a dose so
1001 low that it presents negligible safety concerns from carcinogenic and noncarcinogenic toxic
1002 effects.

1003
1004 **Secondary packaging component:** A packaging component that is in direct contact with a
1005 primary packaging component and may provide additional protection for the drug (e.g., labels,
1006 adhesives, over-pouches, wraps, cartons).